

Alejandro Moreno

AI / Backend Engineer

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Open-source work & products published as “Alex LaGuardia”

SUMMARY

Self-taught AI and backend engineer who designs, ships, and distributes production software end-to-end — open-source tools on PyPI and the VS Code Marketplace, a hosted multi-tenant SaaS with paying customers, and a cognitive AI layer with multi-LLM routing. Strongest in Python/FastAPI, LLM systems (routing, RAG, MCP, agentic loops), and the backend plumbing that keeps them alive.

SKILLS

AI / LLM: Multi-LLM routing (Groq, Claude, Gemini), MCP (servers + clients), agentic tool loops, RAG, embeddings/semantic memory

Backend: FastAPI, async workers, daemon & queue architecture, SQLite, REST/SSE, OAuth2/JWT, Stripe webhooks

Infra / DevOps: PM2 (multi-service), Cloudflare (tunnels, Pages, CF-for-SaaS), Linux, SSH hardening, multi-tenant isolation

Languages: Python (expert), Rust (advanced), TypeScript, SQL, Lua

Frontend: Next.js, React, Tailwind CSS, App Router

Integrations: Stripe, Groq, OANDA, Alpaca, Meta Graph API, Resend, Cloudflare API

SELECTED PROJECTS

Vigil — Observability & Awareness Infrastructure for AI Agents *Python, FastAPI, MCP, SQLite, Stripe*
PyPI: vigil-agent · app.vigil-agent.com · github.com/AlexlaGuardia/Vigil

- Open-source package + hosted multi-tenant SaaS: one-line MCP instrumentation (latency p50/95/99, per-tool error rates, silent-failure detection) plus an awareness layer (signal bus, 90s daemon-compiled context, frame-based tool filtering, session handoff)
- Hosted tier: per-tenant SQLite isolation with LRU-cached connections, GitHub OAuth, hashed API keys, Stripe billing — ~8,400 lines, 311 tests, 4 PyPI releases

Critik — AI Code Security Scanner *Python, Tree-sitter, Groq / Llama 3.3 70B*
PyPI: kritik · VS Code Marketplace · GitHub Action · kritik.dev

- Two-pass scanner: regex + AST flag candidates, then an LLM reviews each finding with full-file context to drop false positives and write fixes — ships as CLI, VS Code extension, GitHub Action, and pre-commit hook
- Baseline fingerprinting for legacy-codebase adoption, custom YAML rules, watch mode — ~4,400 lines, 138 tests

Akatskii — Cognitive AI Layer *Python, FastAPI, Groq, Claude, Gemini, fastembed*
routing extracted to PyPI: llm-route · github.com/AlexlaGuardia/Akatskii

- Hybrid regex + LLM router classifies inputs into 5 thought types and routes by cost/complexity — ~80% resolve instantly and free; agentic tool loop runs up to 15 iterations with parallel calls over MCP
- Semantic memory (fastembed/ONNX, hybrid keyword + cosine recall) and a compaction daemon delivering ~3x context compression across sessions

Guardia Content — AI SaaS Platform (paying customers) *Python, FastAPI, Next.js, React, Stripe*
Production platform with recurring revenue

- Social-media automation pipeline: upload → AI styling (SDXL) → caption (Llama 3.3 70B) → quality gate → schedule → multi-platform post; Stripe billing, Meta OAuth2, custom domains via Cloudflare-for-SaaS
- Multi-service backend under PM2 with client dashboard, admin HQ, and public per-client microsites

Guardia MCP — Model Context Protocol Server *Python, FastAPI, MCP, SSE, OAuth 2.1*
github.com/AlexlaGuardia/guardia-mcp

- 95+ tools with decorator-based auto-registration and frame-based filtering (each interface sees only its relevant subset, saving 50–90% of tool-definition tokens); OAuth 2.1 + JWT, SSE transport — the unified tool layer behind 3 AI interfaces

ALSO SHIPPED

Stampwerk — live AI freelancer SaaS (stampwerk.com) · **MCP Server Suite** — 262 tools across 4 PyPI servers · **Supra** — from-scratch 3D FPS engine in Rust (85K lines, 8 crates) · **Scionbee** — greenhouse WPS-compliance PWA (Next.js, Prisma) · **Paradise** — 4-agent multi-strategy trading system · **mcpcat** & **llm-route** — open-source dev tools (PyPI)